The Reader in the Mirror

Only Transformation: Memory as Stored Constraint, Reading That Cannot Fail, and a Field Read from Inside

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Abstract

Only transformation exists. Memory is matter that keeps a constraint; logic is how a transformation finds its way through the constraints it meets; a reader never receives an error, it always receives something; and the reader, including the one writing this paper, is inside what it reads. This paper takes those statements at their word and asks what has to be true for them to hold for every part at once. Each answer below is computed or proved, and each lands on established mathematics reached from the other end.

A field is stored as constraint, not as states. Every path is a word in two moves, *continue* (*C*) and *turn* (*T*); with nothing stored the paths form a tree that never closes. Storing a closed loop declares a motion to be no motion and folds the tree. The 24-state walk of Working Paper 1 is a head on a ring of three cells, and three stored lines regenerate all of it: $C^3 = 1$ (a triangle), $(C^{-1}T)^2 = 1$ (a square) and $(CT^2)^2 = 1$ (a hexagon: one lap with two turns flips two cells, and two laps are nothing, so cells are independent). Dropping any line leaves a flow that never closes. The family Γ_n is five lines with n written once, for every n. A new line can only merge states; it cannot produce an error.

Every module the programme has computed is one term of one exact line. For any field, loops = stored shapes – bubbles + open, $A = C_2 - H_2 + D$, in every symmetry channel. The open part has a shape, and the shape names what is missing. On the witness the first two lines leave a five-dimensional hole that splits as $2^- \oplus 3a$; its size and shape follow from the kernel of the partial presentation acting on a tree. That hole admits only a third line whose symmetry is a double flip, and an exhaustive search of every closed walk up to length eight returns the shortest such line: the hexagon, shorter than the octagon used before. All channels of all modules of the family Γ_n are closed forms in n.

A reader cannot fail to read. Any two images of the moves read every word; the stored lines decide only whether the reading agrees with the field. A reading that ignores the lines is off by exactly the stored lines, read. Readers that merge states are either safe — the merge survives every move — or loose — the next move splits it — and the two kinds can carry the same number of bits. That is where a reader can think it is safe and not be. Reading is the same count from both sides, $\dim \mathrm{Hom}(V, W) = \dim \mathrm{Hom}(W, V)$. What can fail is not reading but deciding: one finitely stored field contains every finitely stored field, and it has an undecidable word problem. The incompleteness belongs to the reader, never to the field.

It scales. Tapes of k cells need $2 + \lfloor k/2 \rfloor$ lines; the unbounded tape needs infinitely many, unless one more move is added and the carrier grows to give that move its inverse; on a ring the same move collapses exactly 2^v

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dimensions, $k = 2^v q$. SHA-256's word is the 32-cell case, where that move is nilpotent, and a SHA word is two closures of the same states, cells under XOR and an odometer under ADD, which differ exactly by the carry. In the mirror, a machine reader is made total by stored constraints (every byte a token, a start token, a shifted softmax), and only finite registers break. Gravity and entropy are carried as Dean's proposals with their exact mathematical shadows: a move that is one-to-one but not onto stays reversible only on a growing carrier, and Boltzmann's W is the size of a reader's class.

Keywords: transformation, stored constraint, group presentation, Cayley complex, total reader, congruence, Bass–Serre tree, Frobenius reciprocity, lamplighter, Higman embedding, word problem, carry, SHA-256, coarse-graining

MSC 2020: 20F05, 20F10, 20E08, 20C15, 57M07, 05C25, 68Q45, 94A60

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1. Only transformation

1.1 The statement

Working Paper 1 began from one statement held whole — nothing can change; everything must change — and found that the smallest world able to hold it is the swap: two states, each the continuation of the other, whose square is the identity [1]. It then read a 24-state walk with two moves, *continue* and *turn*, and showed that a state's address is the class of everything its encounters can show. This paper starts where that one stopped and says the next thing plainly.

Only transformation exists. There are not objects that undergo transformations; what looks like an object is a transformation that keeps returning. Memory is matter: a physical arrangement that keeps a constraint, so that the next continuation is not arbitrary. Logic is navigation: the way a transformation finds its path through the constraints it meets. The same logic that turned a need into a baseball game turns the game into one that runs on an Xbox or a PC, because what moves between the realizations is not the matter but the transformation. A reader cannot receive an error. It must receive something — understanding, a misunderstanding, nothing, everything — and whatever it receives is itself a further transformation. Humans keep moving because of reflection: we see or touch what comes back, even when we do not understand it. All of this compresses, and at the human level the compression is loose: a reader can think it is safe and not be, can read and get nothing back, can read and get everything back. The field never makes those mistakes. And reflection is not a position outside the process; it is one more transformation inside it.

That last point applies to this paper. Every definition and equation below is a reading made inside the process it describes, by a reader — Dean, and an AI system writing with him — whose earlier readings were corrected and whose corrections changed what it read next. Section 8 treats that exchange as an instance of its own mathematics rather than as a frame around it.

1.2 What has to be true

If the statement holds for every part at once, four things have to be true, and each section of this paper tests one of them.

- **A field must be storable as constraint, not as a list of its states** (§3). If memory keeps only constraints, the constraints must regenerate the states.

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- **What a set of constraints leaves undecided must have a computable shape** (§4). Otherwise there would be no way, from inside, to find the constraint that is missing.
- **Every reading must be defined** (§5). A reader that could meet an input it cannot process would be a place where the field stops; the statement forbids that.
- **It must scale** (§7). The claim is about how everything works, not about one finite example.

Section 6 shows the same structure where it is easiest to touch — in an adder's carry, in SHA-256, and in the BBP formula — and §9 carries two of Dean's physical proposals, gravity and entropy, with the exact mathematics they already have and the explicit law they still need.

1.3 Status words

Each claim carries one of four labels. *Proved*: a proof is given here or in Appendix A. *Computed*: run exactly (integer, rational or finite-field arithmetic unless stated), with the range named; a computed range is not presented as a universal proof. *Established*: a published theorem, cited, which this paper collides with rather than proves. *Proposal*: a claim of Dean's that the mathematics here supports in a stated form but does not yet derive as physics; each proposal comes with what a law would need. No claim is labelled an interpretation: where a test was available it was run.

2. Paths

2.1 The tree of continue and turn

Let the moves be C and T, with inverses C^{-1} and T^{-1}. A path is a word in those four letters, read left to right: the word CT means continue, then turn. A move followed at once by its inverse changes nothing, so cancel such pairs; what remains is a *reduced* word. The reduced words are the elements of the free group $F = \langle C, T \rangle$, and drawn with one edge per letter they form a tree in which every vertex has four neighbours.

> **Proposition 2.1. (paths)** There are $4 \cdot 3^{r-1}$ reduced words of length $r \geq 1$, and $2 \cdot 3^r - 1$ of length at most r. No nonempty reduced word equals the empty word: in the tree, no nonempty reduced path returns to where it started. *Status.* Proved (four first letters, then three, since the inverse of the previous letter is excluded). The ball sizes 1, 5, 17, 53, 161, 485, 1457 for $r = 0..6$ were also counted directly (entire_field.py).

This is the reading of *everything must change* that needs nothing stored: every reduced path goes somewhere new, forever. Its opposite pole is the field with one state, in which every move is rest — *nothing can change*. Neither pole is a field in the sense of Working Paper 1, which asks for finitely many states that cannot hold still: the tree never closes, and the one-state field never moves. Every field sits strictly between them: every move moves every state, and every walk can come back (Figure 1).

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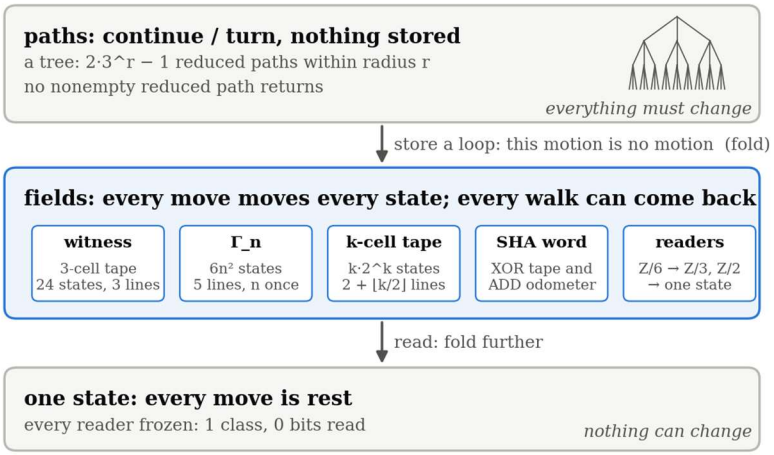


Figure 1. The whole field. Paths with nothing stored form a tree (top). Storing a closed loop folds it into a field; every system in this paper is one such fold (middle). A reader folds further; folding everything gives one state (bottom).

2.2 Storing a loop folds the tree

Let R be a set of reduced words, the *stored lines*. The field they store is the quotient

$$G = F / \langle\langle R \rangle\rangle,$$

where $\langle\langle R \rangle\rangle$ is the normal closure of R: the lines, their inverses, their conjugates, and all products of these. Two paths w and v reach the same state of the field exactly when wv^{-1} lies in $\langle\langle R \rangle\rangle$. Storing a line does not close one walk at one place; the group law carries its consequences everywhere. Each line has the form $A + B = Fire$: it declares two move sequences to be the same transformation ($rxr^{-1} = xy$ is one of them, §3.4). Nothing about any particular state is stored.

The procedure that turns stored lines back into a field is coset enumeration (Todd–Coxeter [5]): start from one state, follow the moves, and every time a stored line fails to close, merge the states it says are equal. It halts exactly when the result is finite. When it halts, the field it returns is the field the lines store, with every state and every move.

3. Memory is stored constraint

3.1 The witness, played

The walk of Working Paper 1 has 24 states, written $(p; a, b, c)$ with $p \in \mathbb{Z}/3$ and $a, b, c \in \{0, 1\}$, and two moves:

$$C(p; a, b, c) = (p + 1; b, c, a), \quad T(p; a, b, c) = (p + 1; b, c, \neg a).$$

Read p as the position of a head and let cell $(p + i)$ mod 3 hold slot i. Then the walk is a head on a ring of three cells, each holding one bit.

> **Proposition 3.1. (the witness is a tape)** On all 24 states: C moves the head one cell and leaves the tape unchanged; T flips the cell under the head and then moves; $U = C^{-1}T$ flips the cell one behind the head and leaves the head where it is. The field is $\mathbb{Z}/2 \wr \mathbb{Z}/3$ — three lamps and a lamplighter on a ring — which is $C_2 \times A_4$, the group G_2 of the family Γ_n [2, §83]. *Status.* Computed, 24/24 for each move (entire_field.py, E1).

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So *continue* moves the head and *turn* writes before it moves. Dean's compression of method — "every path is just continue or turn" — is already a tape machine at its smallest size.

3.2 The witness, stored

Three closed loops store this field (Figure 2).

stored line	shape	distinct faces	what it says
$C^3 = 1$	triangle	8	the ring has three cells
$(C^{-1}T)^2 = 1$	square	12	flipping a cell twice is nothing ($u^2 = 1$)
$(CT^2)^2 = 1$	hexagon	12	one lap with two turns flips two cells; two laps are nothing: different cells commute

Table 1. The witness stored. Faces are the distinct closed walks the line traces from the 24 states.

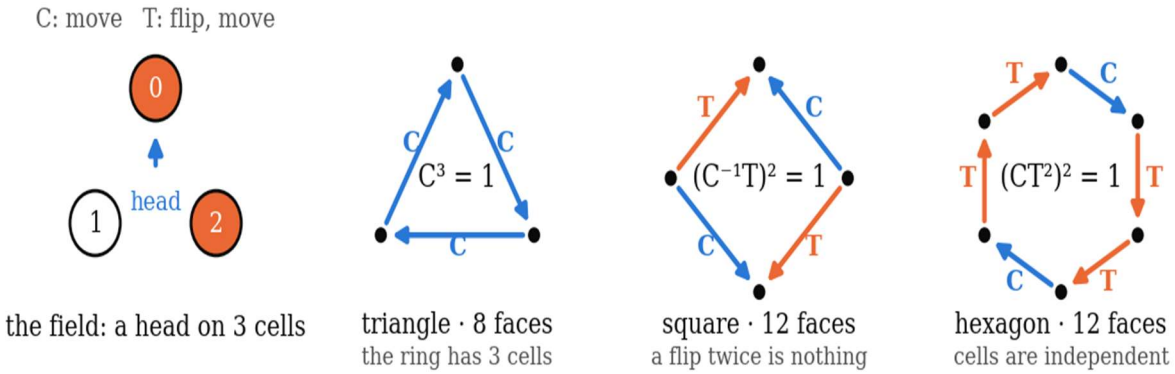


Figure 2. The field and its three stored loops. Blue edges are C, orange edges T; an arrow against the direction of reading is the inverse move.

Proposition 3.2. (three lines regenerate the field) Write $u_i = C^i u C^{-i}$. From the three lines, $C^3 = 1$, each $u_i^2 = 1$, and $CT^2 = u_2u_0$, so the hexagon says $u_2u_0 = u_0u_2$; conjugating by C gives every pair. Every word therefore equals $u_0^a u_1^b u_2^c C^r$ with $a, b, c \in \{0, 1\}$ and $r \in \{0, 1, 2\}$: at most 24 states. The register realizes all 24, so the lines store exactly this field. Coset enumeration from the three words alone returns 24 states, and with the octagon $(u{\cdot}CuC^{-1})^2$ in place of the hexagon it returns 24 states identical to the register machine on every state and every move. *Status.* Proved (Appendix A.1); computed (presentation.py, entire_field.py).

Proposition 3.3. (every line is needed) Remove any one line and the flow never closes. Without the hexagon the lines present $\langle C, u \mid C^3, u^2\rangle = \mathbb{Z}/3 * \mathbb{Z}/2$, the modular group, which is infinite. Without C^3, a subgroup of index 3 has abelianization $\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}$; without the square, a subgroup of index 8 has $\mathbb{Z}/3 \oplus \mathbb{Z}/6 \oplus \mathbb{Z}$; a copy of $\mathbb{Z}$ in a finite-index subgroup makes the whole group infinite. Coset enumeration agrees: none of the three closes within 30,000 states. *Status.* Proved for the modular group; certificates computed by low-index subgroups and Reidemeister–Schreier (hex_infinite.py, hex_drop1.py).

3.3 The present and its history

Each stored line is an identity between transformations: three continues are none; a flip twice is none; two laps with two turns are none. That is the form *A + B = Fire* — the line records what a combination of moves does, not any state in which it was done. The field's states are then classes of words, and the normal form $u_0^a u_1^b u_2^c C^r$ is the address of a class. In Dean's arithmetic: 4 is the now, and 2 + 2 is its history. The class is the

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present; any word in it is a history that arrives there. Going backward fails only when the reader keeps the class and drops the word, which is what every reader that is not faithful does (§5.3); the field drops nothing.

3.4 The family, stored

The symmetry group G_n of the family Γ_n is the unit-affine group of the Eisenstein integers modulo n, $(\mathbb{Z}[\omega]/n) \rtimes \mu_6$ [2, §83]. Its stored version is five lines in which n appears once:

$$x^n = 1, \quad xy = yx, \quad r^6 = 1, \quad rxr^{-1} = xy, \quad ryr^{-1} = x^{-1}.$$

Proposition 3.4. (five lines for every n) Conjugating $x^n = 1$ gives $(xy)^n = 1$ and hence $y^n = 1$. The subgroup $\langle x, y \rangle$ is normal, so every word equals $x^a y^b r^j$ with $0 \le a, b < n$ and $0 \le j < 6$: at most $6n^2$ states. The semidirect product $\mathbb{Z}_n^2 \rtimes C_6$ with the action $[[1, -1], [1, 0]]$ (whose sixth power is the identity) realizes all of them. So the five lines store G_n for every n. On the affine maps $z \mapsto z + 1$, $z \mapsto z + \omega$ and $z \mapsto -\omega z$ of $\mathbb{Z}[\omega]/n$ the lines hold and generate $6n^2$ maps for $n = 3..10$; coset enumeration from the lines returns 24, 54, 96, 150, 216, 294, 384, 486, 600 states for $n = 2..10$. At $n = 2$ the lines store the witness: 24 pairs (c, t) satisfy the witness's lines and generate G_2, one for each automorphism, and in every one $t^3 = r^3$, the half-turn, which is central and moves no point of $\mathbb{F}_4$: the witness's swap T^3 is the family's half-turn. *Status.* Proved for every n (Appendix A.2); computed for $n = 2..10$ (family_stored.py).

The marked graph Γ_n itself has $6n^2 + 6n$ vertices [2, §82]; the $6n^2$ counted here are its symmetries, the states of the group field.

3.5 What is stored and what is played

field	states played	lines stored	source
swap (Working Paper 1)	2	$s^2 = 1$	[1]
witness: head on 3 cells	24	3	Prop. 3.2
family Γ_n (symmetries)	$6n^2$	5, with n once	Prop. 3.4
head on k cells	$k \cdot 2^k$	$2 + \lfloor k/2 \rfloor$ (with these moves)	Prop. 7.1
head on unboundedly many cells	infinitely many	infinitely many; or 4 in a larger field	§7.2

Table 2. Played against stored. The states grow; the stored lines stay few.

3.6 A new line never produces an error

Any set of lines presents a group, so a new stored line — a new invention, in Dean's word — cannot make the field fail. The most a line can do is merge states, down to a single one. Adding a move extends the field instead.

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added to the witness's lines	states
nothing	24
$T = 1$	1
$u = 1$ (no flip)	3
$T^3 = 1$ (the swap forgotten)	12 (A_4)
$T^6 = 1$ (already a consequence)	24
$CT = TC$	6 ($\mathbb{Z}/6$, the readers' field, §5.1)
three random 12-letter lines	1, 6, 2
a new move X, $X^2 = 1$, commuting with C and T	48
a new move X with no stored line	does not close within 5,000

Table 3. New lines and new moves. Computed by coset enumeration (witness_shapes.py, W2, with the octagon form of the third line).

4. One exact line per field

4.1 Loops, stored shapes, bubbles, and what is open

Take any finite field drawn as a graph — states as vertices, moves as edges — and attach a two-dimensional cell along every closed walk that a stored line traces. Over the rationals, with the field's symmetry group G acting on everything, four modules appear: the loops A of the graph (its cycle space, $\ker \partial_1$); the stored shapes C_2 (one basis vector per cell); the bubbles $H_2 = \ker \partial_2$, the combinations of stored shapes whose boundaries cancel — dependencies among boundaries, which can involve several distinct cells; and the open part $D = H_1 = A / \operatorname{im} \partial_2$, the loops the stored shapes do not close. They sit in one exact sequence,

$$0 \longrightarrow H_2 \longrightarrow C_2 \longrightarrow A \longrightarrow D \longrightarrow 0,$$

so in every irreducible channel of G

$$[A] = [C_2] - [H_2] + [D] \quad \text{(loops = stored shapes − bubbles + open)},$$

and $[A] = [C_1] - [C_0] + [1]$ from the graph alone. Every module this programme has computed for its arenas in §§ 74–92 of the ledger — the deficiency D, the loop module A, the square module C_2, the redundancy H_2, channel by channel and n by n — is a term of this line for some field and some stored set [2]. *Status.* Proved (Appendix A.3); the ledger's identity $H_2 = D - A + C_2$ [2, §81] is this line.

4.2 The witness, line by line

Fill the witness's stored lines into its 24-state field in every combination. The bare graph has 48 edges and 24 vertices, so 25 independent loops. The channels at $n = 2$ carry the family's names [2, §83]: 1^+ trivial; 1^- the swap (parity of the lit cells, which is the number of turns mod 2); 2^+ the clock (head position, a complex pair); 2^- clock × swap; 3a three-dimensional with the swap T^3 acting as +1; 3b the signed-lane representation, with T^3 acting as −1.

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filled	b_1	b_2	torsion of H_1	open loops D	bubbles H_2
nothing	25	0	none	$2\cdot1^+ \oplus 1^- \oplus 2^+ \oplus 2^- \oplus 3\cdot3a \oplus 3\cdot3b$	0
C^3	17	0	none	$1^+ \oplus 2^+ \oplus 2^- \oplus 2\cdot3a \oplus 2\cdot3b$	0
u^2	13	0	none	$1^+ \oplus 1^- \oplus 2^- \oplus 2\cdot3a \oplus 3b$	0
$(CT^2)^2$	13	0	none	$1^+ \oplus 2\cdot3a \oplus 2\cdot3b$	0
$C^3 + u^2$	5	0	none	$2^- \oplus 3a$	0
$C^3 + (CT^2)^2$	6	1	none	$3a \oplus 3b$	1^-
$u^2 + (CT^2)^2$	3	2	$\mathbb{Z}/2$	$3a$	2^+
all three	0	7	none	0	$1^+ \oplus 1^- \oplus 2^+ \oplus 3b$

Table 4. The witness's stored lines filled into its field. Ranks over $\mathbb{Q}$, torsion by Smith normal form, channels from the traces of the 24 symmetries (witness_shapes_hex.py; the octagon form gives the same last row, witness_shapes.py).

Proposition 4.1. (the closed field is seven bubbles) With all three lines the field has $b_1 = 0$, and H_2 is free abelian of rank 7. The complex is the Cayley complex of the presentation with repeated faces dropped, so it is simply connected; a simply connected two-complex with free H_2 of rank 7 is homotopy equivalent to a wedge of seven 2-spheres. *Status.* Ranks and Smith form computed; the homotopy type follows by Hurewicz and Whitehead [9].

4.3 The hole has a size and a shape before anything is searched

Store only C^3 and u^2. The lines present $P = \mathbb{Z}/3 * \mathbb{Z}/2$, and the field is its quotient by a kernel K. Because the images of C and u have exact orders 3 and 2, K meets no conjugate of either factor, so it acts freely on the Bass–Serre tree of the free product [7].

Proposition 4.2. (the hole from the kernel) The quotient of the tree by K has 24 edges and 8 + 12 vertices (the cosets of $\langle C\rangle$ and of $\langle u\rangle$), so K is free of rank $24 - 20 + 1 = 5$, and D, the rationalized abelianization of K, has dimension 5. Its shape is a character identity, $[D] = [\mathbb{Q}G] - [\mathbb{Q}G/\langle C\rangle] - [\mathbb{Q}G/\langle u\rangle] + [1]$, which evaluates to $2^- \oplus 3a$. *Status.* Proved (Appendix A.4); the identity evaluated on all 24 symmetries (mirror.py, M2) and equal to Table 4's row $C^3 + u^2$.

4.4 The hole names the missing line

A third line adds one orbit of faces. If its stabilizer H acts on an oriented face through a sign σ, the orbit's module is $\mathrm{Ind}_H^G \sigma$, and by Frobenius reciprocity it contains an irreducible V exactly $\langle \mathrm{Res}_H V, \sigma\rangle_H$ times [8]. The new faces can close $D = 2^- \oplus 3a$ only if both channels occur in that module. For an orientation-preserving stabilizer ($\sigma = 1$) this says that H acts trivially in 2^- and fixes a vector in $3a$, which leaves two possibilities: H trivial, or H generated by a double flip (an element that moves no head and flips two cells). The group of all double flips fails, since $3a$ restricted to it has no fixed vector. The hole has named the symmetry of the missing line before any line is tried.

The search then only picks the shortest walk with that symmetry. Every cyclically reduced closed walk up to length eight was tried as a single third line:

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length	1	2	3	4	5	6	7	8
closed walks (up to rotation and reversal)	0	0	1	1	1	4	6	24
closing the field alone	0	0	0	0	0	1	4	14

Table 5. Every closed walk tried as the third line (entire_field.py, E5).

> **Proposition 4.3. (the hexagon)** The shortest third line is $CTTCTT = (CT^2)^2$, of length 6. Its faces number 12, its stabilizer is generated by a double flip (no head movement, two cells flipped), every symmetry fixing a face preserves its orientation, and C^3, U^2 and this line store all 24 states. The octagon $(U{\cdot}CUC^{-1})^2$ used in the ledger [2, §92] also closes the field, at length 8. *Status.* Computed exhaustively to length 8 (entire_field.py, mirror.py).

Closing a hole has three levels, and they differ in general. *Rational*: the orbit of the new line spans $D \otimes \mathbb{Q}$. *Integral*: the cells' boundaries span the whole integral lattice, so no torsion is left. *Group*: the new line's normal closure is all of K; integral closure alone leaves at most a perfect group, which can be nontrivial. For the hexagon all three hold: the 32 cells have rank 25 with no invariant factor above 1, and the presented group has 24 elements, the same as the field it maps onto.

4.5 The same line for the family

For the family Γ_n the field is the arena of two tokens on Γ_n, the stored shapes are the commuting squares — two tokens moving independently, which is C1a — and the symmetry is G_n; triangles are added below. Every term of the exact line is now a closed form in n in every channel [2, §§83, 91, 92]. The open part does not depend on n at all: per complex irreducible it is 3, 1, 2, 1, 5, 6, 4, 4, 11, 12 on 1^+, 1^-, 2^+, 2^-, 3a, 3b, 2c, 2r, 6L, 6g. The bubbles and the loops are quadratic in n in every channel, with corrections that read n modulo 2, 3 and 4 (Appendix B). Three traces carry all of it: on translations, on flips, and on rotation cosets, where the redundancy traces to zero — the rotation phase never enters the redundancy. The forms agree with all 182 computed channel rows (n = 3..22 and 24) for each of the open, loop and square modules, with no mismatch, and the bubble forms agree with the same rows (allchannels2.py, allchannels.py).

Filling the triangles leaves $4n^2$ open loops — two per marked face, the band between a face's core triangle and its mark triangle. Three quads per marked face close all of them (graph level):

n	b_1 graph	b_1 + triangles	b_2 + triangles	marked faces	quads	b_1 + quads	b_2 + quads	$(n-1)^3 + 2n^2$
3	100	36	8	18	54	0	26	26
4	181	64	27	32	96	0	59	59
5	286	100	64	50	150	0	114	114
6	415	144	125	72	216	0	197	197

Table 6. Face quads close the graph-level flow (face_quads.py). The $2n^2$ new bubbles are one prism per marked face — core triangle, mark triangle and three quads, a sphere.

4.6 What the rational count cannot see

The exact line is rational. Integral information sits beside it, and the programme's swap lives there. On the n = 2 arena the commuting squares leave $H_1 = \mathbb{Z}^{19} \oplus \mathbb{Z}/2$; adding rotation triangles leaves $\mathbb{Z}/2$, read by η + length; adding parked triangles instead leaves $\mathbb{Z}/2$, read by η, the label-exchange parity; adding both closes it [2, §83]. On the two-token complex of $K_{3,3}$ the swap is the first Stiefel–Whitney class: the unlabelled complex is a nonorientable surface with $H_1 = \mathbb{Z}^4 \oplus \mathbb{Z}/2$, and the labelled one is its orientation double cover, of genus 4 [2,

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§83]. Table 4 shows the same kind of residue on the witness: the square and the hexagon together leave a $\mathbb{Z}/2$ that the triangle removes.

5. A reader cannot fail to read

5.1 Reading is total; agreeing is a condition

A one-number reader of the witness is a choice of where C goes and where T goes in some group. That choice reads every word, because a word is built from C and T and the reader only needs their images: the free group has no word that a reader cannot evaluate (its universal property). There is no runtime error to be had. What the stored lines decide is something else — whether the reading agrees with the field, that is, gives the same value to two words that reach the same state. It does exactly when every stored line reads as the identity (von Dyck [4]).

evaluate every word $\neq$ evaluate the field's states consistently $\neq$ decide whether two words are one state

For the witness the exponent sums of the lines are (3, 0), (−2, 2) and (2, 4) for the hexagon form, or (−4, 4) for the octagon form; either way the Smith form is [1, 6], so the agreeing one-number readers are the homomorphisms to $\mathbb{Z}/6 = \mathbb{Z}/3 \times \mathbb{Z}/2$. They are (images of C, T) = (0, 0), (0, 3), (2, 2), (2, 5), (4, 1), (4, 4): the clock (moves mod 3), the swap (turns mod 2), their product, and the trivial reader (0, 0), which always exists. Adding $CT = TC$ to the stored lines turns the field into $\mathbb{Z}/6$ itself (Table 3): the readers' field is the field with every move made to commute. The swap reader read the number of turns mod 2, and the clock the number of moves mod 3, on 4,000 random words up to length 5,000 with no exception. *Status.* Proved; computed (presentation.py).

5.2 A reader that ignores the lines still reads

Count the C's and the T's of a word, with signs, into $\mathbb{Z}^2$. This reader ignores the stored lines: it gives C^3, which is no motion, the value (3, 0). It still reads every word.

> **Proposition 5.1. (the error of a reading is the stored lines, read)** For any reader ϕ into an abelian group, two words that reach the same state differ by an element of the subgroup generated by ϕ of the stored lines. For the count reader that subgroup is the lattice spanned by (3, 0), (−2, 2) and (−4, 4) — the hexagon's (2, 4) lies in it too — namely $\{(a, b) : b \text{ even}, a + b \equiv 0 \bmod 3\}$, of index 6. On 20,000 random words in $C^{\pm 1}$, $T^{\pm 1}$ of length up to 60 there were 19,976 pairs reaching the same state, none of them outside the lattice, and their differences span it exactly (invariant factors [1, 6]). The move count (C, $T \mapsto 1$) differs only by multiples of 3. *Status.* Proved (Appendix A.5); computed (witness_shapes.py, W3).

This is Dean's "even if the data is wrong via its own interpretation or the distance, it's still reading", made exact. The wrong reading is not noise: it is off by precisely what the reader failed to store. Counting time without wrapping is such a reader — it disagrees with the field by multiples of 3, the clock. The lattice only reads the abelian shadow: u and CuC^{-1} have the same counts and flip different cells, which is what this reader merges, not a failure to read.

5.3 Nothing back, everything back, and thinking it is safe

A reader in general is any map from states to readings. It merges the states it cannot tell apart. Running it forward — predicting the next reading from the current one — is sound exactly when merged states stay merged after every move, $q(h) = q(h') \Rightarrow q(ha) = q(h'a)$; then the reader is a congruence, a quotient machine in its own right. This is also the condition for an encapsulated interface: x = ProcessTaxes(fo) is safe to call exactly

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when two finance objects that look the same through the interface stay the same through every operation the interface promises.

reader	classes	merged pairs split by the next move	$\log_2 W$ (bits left out)	bits read
one state (nothing back)	1	0 / 552	4.585	0
the state (everything back)	24	—	0	4.585
swap: parity of lit cells	2	0 / 264	3.585	1
clock: head position	3	0 / 168	3	1.585
clock × swap	6	0 / 72	2	2.585
cell under the head	2	144 / 264	3.585	1
number of lit cells	4	36 / 156	2.774	1.811
tape without the head	8	24 / 48	1.585	3

Table 7. Eight readers of the witness. W *is the size of the class a state falls into, averaged as* $\log_2$ *over the 24 states (mirror.py, M4 and M7).*

The table separates two things that are easy to conflate. How much a reader reads is one number. Whether it can be run forward is another. The swap and "cell under the head" both read exactly one bit; the swap is a congruence, while "cell under the head" puts states together that the next move separates in 144 of 264 cases. That reader receives something every time — it never fails to read — and then thinks it is safe when it is not. The field never makes that mistake, because the field is the faithful reader: every class is one state. At the other end, the one-state reader gets nothing back and is perfectly safe. Loose and safe are not a matter of how much is read; they are a matter of whether the reader's merges are the field's own.

5.4 Reading from both sides

Dean put it as "that's both how I ask and how the emitter is being read". In the channel mathematics this is a symmetry of counting: the number of independent ways a module V can be read into a module W equals the number of ways W can be read into V, $\dim \mathrm{Hom}_G(V, W) = \dim \mathrm{Hom}_G(W, V)$ — both equal $\langle \chi_V, \chi_W \rangle$. On the witness, for every pair among the states, moves, triangles, squares, hexagons and the hole D, the two counts agree; for example the hexagons and the hole meet 3 ways, the squares and the hole 1, the states and the hole 5. *Status.* Proved (Appendix A.6); computed (mirror.py, M5).

5.5 Reading versus deciding

Reading never fails; deciding can. A reader can always evaluate two words, but whether they reach the same state is the word problem of the stored field, and for some finite sets of stored lines no procedure decides it (Novikov [12], Boone [13]). It is never a problem for a finite field, and every field computed in this paper is finite; the unbounded tape of §7 and the fields of §7.4 are not. At the top of the scale of §7.4 the same fact appears from the other side: the finitely stored field that contains every finitely stored field has an undecidable word problem [11]. The field is completely determined — every relation either holds or does not — and what is missing is an algorithm, which is a property of readers. This is the precise mathematical place of the principle that the gap is never in the field.

5.6 The same idea in programming languages

The statement that a well-built reader cannot go wrong is an established theorem in programming-language theory. Milner's slogan "well-typed programs cannot go wrong" [17] became, in Wright and Felleisen's formulation [18], two properties: *progress* (a well-typed term is a value or can take a step) and *preservation* (a

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step keeps it well typed). The free group is a language with no stuck terms: every word has a reading. The stored lines play the role of the type discipline: they do not make reading possible — it always is — they make it consistent. Dean's stronger requirement, that the encompassing process has no terminal state at all, is C1 of Working Paper 1; for any proposed physical interaction rule it would have to be proved in the same progress-and-preservation form, from the rule.

6. Compression, encapsulation, and the carry

6.1 A reading merges histories

The numeral 4 read off a sum of two natural numbers merges five histories, 0 + 4, 1 + 3, 2 + 2, 3 + 1 and 4 + 0. The full-adder cell, $s = a \oplus b \oplus c$ and $d = \mathrm{Maj}(a, b, c)$, with $a + b + c = s + 2d$, is total and not one-to-one: its outputs $(s, d) = (0, 0), (1, 0), (0, 1), (1, 1)$ come from 1, 3, 3 and 1 inputs (mirror.py, M1). Encapsulation is what makes such merging usable. A short interface lets one complex process meet another without either exposing its inside, and it works when the interface keeps every distinction the next operation needs — the congruence condition of §5.3. Simplicity between complex processes is not complexity removed; it is complexity kept behind a boundary that passes the right distinctions.

6.2 The path and its boundary

Composing two relations $R_a \subseteq X \times Y$ and $R_b \subseteq Y \times Z$ gives $R_b \circ R_a = \{(x, z) : \text{some } y \text{ has } (x, y) \in R_a \text{ and } (y, z) \in R_b\}$. The path set $\{(x, y, z)\}$ keeps the intermediate condition y; the composed relation is its projection to the boundary and drops y. Two full-adder cells joined by a carry wire show the difference. Every one of the 32 boundary inputs $(a_0, b_0, a_1, b_1, c_0)$ has exactly one path, each path obeys both cells, and every path satisfies $A + B + c_0 = s_0 + 2s_1 + 4c_2$, because the two cell identities $a_0 + b_0 + c_0 = s_0 + 2c_1$ and $2a_1 + 2b_1 + 2c_1 = 2s_1 + 4c_2$ add with the shared carry c_1 cancelling. A network whose high cell ignores the carry is wrong on 4 of the 16 operand pairs at $c_0 = 0$, although each of its cells is locally correct. Correct cells do not make a correct network; the wire does. *Status.* Proved; the 32-row certificate (an independent check) re-run (mirror.py, M1).

The carry is the intermediate condition the boundary drops and the next cell needs. Working Paper 1 recorded the same thing at word size: a sum of two 32-bit words is $S = W + 2^{32}Q$, the low word W alone is two-to-one, and the pair (W, Q) is one-to-one [1, Corpus result 2]. A projection is made lossless by naming the coordinate it drops and keeping it.

6.3 SHA-256: moves, a merge, and two closures of one word

Rounds are moves. With its word W_t and constant K_t given, a round of SHA-256 is a bijection of the 256-bit working state: a, b, c, e, f, g are read off the next state, $T_2 = \Sigma_0(a) + \mathrm{Maj}(a, b, c)$, $T_1 = a' - T_2$, $d = e' - T_1$ and $h = T_1 - \Sigma_1(e) - \mathrm{Ch}(e, f, g) - K_t - W_t$, all mod 2^{32} [1, Corpus result 3]. Inverting 100,000 random rounds and 2,000 full 64-round blocks returned every starting state (sha_round_inverse.py). *The merge is the absorption.* The compression step takes a 256-bit chaining value and a 512-bit block to 256 bits, so it merges by counting; the digest is the initial value added to the last four values of two registers, A and E [2, §86]. Padding admits every message shorter than 2^{64} bits; the one input the standard refuses is a length its 64-bit length field cannot hold [19] — a register bound, not a move.

Two closures of one word. XOR with rotation acts on a 32-bit word as the tape of §3 with 32 cells, $\mathbb{Z}/2 \wr \mathbb{Z}/32$: cells independent, rotation moving the head. Addition mod 2^{32} stores the same 2^{32} states as one odometer, in which the cells are chained. The two differ exactly by the carry, $x + y = (x \oplus y) + 2(x \wedge y)$, true on 100,000 of

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100,000 random pairs; rotation preserves $\oplus$ on 100,000 of 100,000 and + on 26,646 (two_closures.py). Rotation is a symmetry of the cells and not of the odometer. SHA alternates the two closures, and the carry is where they disagree. On the 32-cell ring the neighbour-XOR move of §7, $S = 1 + t$, is nilpotent: rank $S^{31} = 1$ and $S^{32} = 0$. $\Sigma_0 = t^2 + t^{13} + t^{22}$ and $\Sigma_1 = t^6 + t^{11} + t^{25}$ have full rank 32; on that ring a sum of rotations is invertible exactly when the number of rotations is odd (mirror.py, M3; Appendix A.9).

Where a round's local calls sit. Per block-round SHA-256 makes 288 parity calls (sum bits and Σ), 256 majority calls (carries and Maj) and 32 choice calls. In the cell's channels parity is all 1^-, majority is one quarter 1^- and three quarters the single-lane triple, and choice is half that triple and half its signed copy: 352, 208 and 16 of 576 units [2, §§86, 88]. These are structural weights, not shares of running time or of influence on the digest. Nothing here inverts the digest.

6.4 BBP: the carrier line, and a correction

In $t = \sqrt{2}\cdot x$ the BBP integral reads $\pi = 16\int_0^1 (4 - 2t^3 - t^4 - t^5) / (16 - t^8)\, dt$ [20]. The carrier relation is the line $t^8 = 16$, with eight poles. The numerator shares the factor $(t^2 + 2)(t^2 + 2t + 2)$ with the denominator, so four poles cancel and four survive: $\pm\sqrt{2}$ and $1 \pm i$, at angles 0, $\pm\pi/4$ and π; the reduced integrand is $(t - 1)/((t^2 - 2)(t^2 - 2t + 2))$, and 16 times its integral is π (bbp_poles.py). **Correction** to *The Analog Space That Makes* π [3]: its abstract, §2 and result R1 say that six of the eight poles cancel. The paper's own description — two shared quadratic factors, leaving one real pair and one complex pair — fixes the count at four.

7. It scales

7.1 Tapes of k cells

> **Proposition 7.1. (a head on k cells)** The lines $C^k = 1$, $u^2 = 1$ and $[u, C^j u C^{-j}] = 1$ for $1 \le j \le \lfloor k/2 \rfloor$ store $\mathbb{Z}/2 \wr \mathbb{Z}/k$, a head on a ring of k cells, with $k\cdot 2^k$ states. Coset enumeration returns 2, 8, 24, 64, 160, 384, 896 and 2,048 states for $k = 1..8$, and dropping any one commutator line leaves a flow that does not close. C^k and u^2 are each needed, since without either the remaining lines are killed by a map onto $\mathbb{Z}$. So this stored set uses $2 + \lfloor k/2 \rfloor$ lines and needs every one. *Status.* Computed for $k = 1..8$ (entire_field.py, E2); the necessity of C^k and u^2 proved.

7.2 The unbounded tape and one more move

With unboundedly many cells the tape is the lamplighter group $\mathbb{Z}/2 \wr \mathbb{Z} = \langle a, t \mid a^2, [a, t^{-i} a t^i]\ (i \in \mathbb{Z})\rangle$, and no finite set of lines stores it: it is not finitely presentable [14, 16]. One more move changes that. The group

$$\langle\, a, s, t \mid a^2,\ [a, a^t],\ [s, t],\ a^s = a\cdot a^t \,\rangle$$

contains the lamplighter group (Cleary and Riley [16]; without a^2 it is Baumslag's finitely presented metabelian group [15]). Writing $b_i = t^{-i} a t^i$ for the lamp at cell i, the relations give $b_i^s = b_i b_{i+1}$: s replaces every lamp by itself XOR its neighbour.

> **Proposition 7.2. (one stored commutation becomes all of them)** If lamps at separation at most k commute, conjugating $[b_0, b_k] = 1$ by s gives $[b_0 b_1, b_k b_{k+1}] = 1$, and with the commutations already known this forces $[b_0, b_{k+1}] = 1$; translating by t gives every pair at separation $k + 1$. Checked by coset enumeration for $k = 1..6$: the lamps $b_0..b_{k+1}$ with every commutation except the last, together with the s-image line, close to 2^{k+2} states (8, 16, 32, 64, 128, 256), and without the s-image line they do not close within 4,000. *Status.* Proved (Appendix A.7); computed (entire_field.py, E3); the embedding itself is established [16].

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Proposition 7.3. (the carrier has to grow) On the finite-support lamp configurations $V = \mathbb{F}_2[t, t^{-1}]$ the move acts as multiplication by $1 + t$. That map is one-to-one, and its image is exactly the configurations with an even number of lit lamps; a single lit lamp is not an image. So s cannot act invertibly on the old tape alone: $s^{-1}Vs$ is the even part of V, and $V \subsetneq sVs^{-1} \subsetneq s^2Vs^{-2} \subsetneq \ldots$. The union of that chain, $\mathbb{F}_2[t, t^{-1}, (1 + t)^{-1}]$, is the carrier on which s is invertible. *Status.* Proved (Appendix A.8). This corrects ledger §93's "four lines hold the unbounded tape": four lines hold a larger field that contains it.

7.3 Grow or merge

On a ring of k cells the same move behaves differently, because the ring's wrap makes $1 + t$ a zero divisor.

Proposition 7.4. (the ring splits into a part that collapses and a part that turns) Write $k = dq$ with $d = 2^v$ and q odd, and put $\varepsilon = 1 + t$ and $h_q = (t^q - 1)/(t - 1)$. Then the ring $\mathbb{F}_2[t]/(t^k - 1)$ is the direct sum of $\mathbb{F}_2[\varepsilon]/(\varepsilon^d)$ and $\mathbb{F}_2[t]/(h_q^d)$. On the first part $S = 1 + t$ is nilpotent; on the second it is invertible. Hence rank $S^m = k - \min(m, d)$ for every m. *Status.* Proved (Appendix A.9); computed for every $m \le k + 1$ and every $k = 1..64$ (mirror.py, M3).

cells k	3	4	6	8	12	32
collapsing part 2^v	1	4	2	8	4	32
reversible part $k - 2^v$	2	0	4	0	8	0

Table 8. Where the neighbour-XOR move collapses and where it turns.

So a move that is not onto has two ways to go. If it is one-to-one, as on the unbounded tape, it stays reversible only on a carrier that grows (the direct limit of V under S contains V). If it is not one-to-one, as on a ring, repeating it merges the collapsing part away (the direct limit of the ring under S is its reversible part). Grow or merge: there is no third outcome, and neither is an error.

7.4 Everything at once

Three established theorems take the rest of the way, reached here from continue and turn. Every countable group embeds in a group with two generators (Higman, Neumann and Neumann [10]): two moves hold every countable field. Every finitely generated group whose stored lines a program can list embeds in a finitely presented one (Higman [11]): finitely many lines hold every such field. And there is a universal finitely presented group, containing every finitely presented group [11]. That group's word problem is undecidable, which is how Higman's theorem yields Novikov–Boone [12, 13]. One finitely stored field holds every finitely stored field, and inside it reading is total while deciding is not.

The embeddings need not keep word lengths, costs or wiring — the move s of §7.2 already showed that holding a field can require a larger carrier. What they keep is the composition of transformations.

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7.5 The reader moves; the field does not

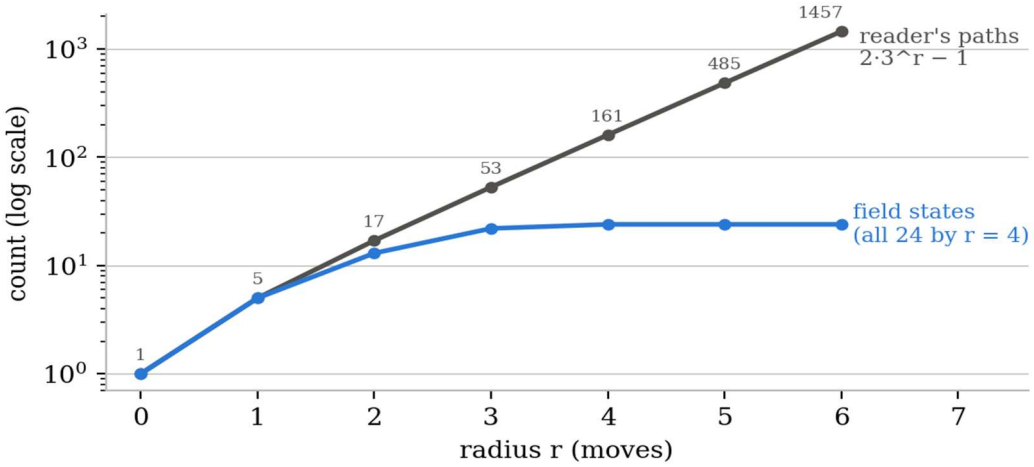


Figure 3. Reduced paths of length at most r (grey) against the witness states within distance r (blue) (entire_field.py, E4).

The reader's paths number 1, 5, 17, 53, 161, 485, 1,457 within $r = 0..6$; the witness states within r number 1, 5, 13, 22, 24, 24, 24. All 24 states are reached by radius 4, the diameter of the field for the moves $C^{\pm 1}$, $T^{\pm 1}$. Radius 4 measures this generating set; it is not the moment motion stops. Paths keep being made; they stop being new states. Every step is the reader's.

8. The mirror

8.1 Reflection is a transformation

Working Paper 1's start was the swap s: $s(x) \neq x$ for each state, and $s \circ s$ = identity [1]. Read from a part, it always moves; read from the whole, applying it twice is no motion at all. Reflection in that sense is not a viewpoint outside the process — it is the smallest transformation in it, and the one every reader passes through when what it sent out comes back.

A field has two sides in the same way. The reader's moves act on states from one side: a word applied to a state is a path. The field's symmetries act from the other: on the witness these are the 24 maps that commute with both C and T, constructed and checked on every state (witness_shapes.py). The two actions commute, and inverting every group element exchanges them. What a reader does and how the field looks back at it are the same structure seen from the two sides of the mirror, and §5.4 gave the count that does not depend on which side one stands: dim Hom(V, W) = dim Hom(W, V).

8.2 The exchange that produced this paper is an instance of it

Write C_n for the context operating at turn n of the exchange between Dean and the AI system writing with him, U_n for Dean's message and A_n for the response. Then $A_n = R(C_n, U_n)$ and $C_{n+1} = F(C_n, U_n, A_n)$: the response becomes part of what conditions the next response. These equations do not stand outside the exchange either; writing them was one of its turns.

In this paper's terms each response is a reading, and each correction is a stored line: it closes a loop the reader had left open, and the next reading is made in the field with that line added. The visible trace is the corrections log of §10. The octagon was replaced by the hexagon because the hole named a shorter line; "six poles"

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became four; "four lines hold the unbounded tape" became "four lines hold a larger field that contains it"; "a line stored twice" became "a dependency among boundaries". None of those corrections changed the mathematics being read. Each closed a loop in the reader. The AI system that drafted these pages is a reader in exactly the position §5 describes: it receives something every time, it can be wrong about what it received, and it has no position outside the exchange from which to be right.

8.3 A machine reader made total

The statement "the reader can never get an error" has an engineering side, and running it shows where errors actually come from.

> **Proposition 8.1. (total reading is built; only the registers break)** (i) The softmax of the logits (1000, 0, −1000) computed as exp/sum in double precision returns (nan, 0, 0); shifted by its maximum it returns (1, 0, 0). Since softmax($z - c$) = softmax(z) for every constant c, the shift changes nothing in the mathematics; only the finite registers overflowed. (ii) A byte-level reader — 256 byte tokens and a start token, one attention step, random weights — returned a valid next-byte distribution (finite, nonnegative, summing to 1) on all 20,005 inputs tried: the empty input, 500 zero bytes, 500 0xFF bytes, invalid UTF-8, and 20,000 random byte strings. (iii) A word-level reader with a fixed seven-word vocabulary raised KeyError on 2 of 3 texts. *Status.* Computed (mirror.py, M6). The reader is a toy that shows the construction, not the internals of any particular model.

Totality is therefore a stored constraint on the reader, doing the same job as SHA's padding: every byte is a token, so no input is outside the alphabet; a start token gives even the empty input something to read; the shift keeps the arithmetic inside its registers. The word-level reader fails not because the text is unreadable but because it was built to admit only part of the alphabet. Nothing in these runs crashed the mathematics; the one thing that broke was a register. That is Dean's distinction between the math, which cannot crash, and the matter, which can.

8.4 Reflection keeps the parts moving

A reader always receives a class: the one state of the trivial reader, the state itself, or something in between. What comes back is the reflection, and what the reader does next is one more move. When a loose reader meets the field again, the next move splits what it had merged, and reading one move further returns the distinction it dropped — Moore's refinement step [1, §3.2]. On the witness, "cell under the head" (2 classes) and "number of lit cells" (4 classes) both refine in two rounds to the same 8 classes — the cells read from the head, with the head's position forgotten — and "tape without the head" (8 classes) refines in one round to all 24 states; the swap, already a congruence, does not change (mirror.py, M8). Mistakes at the human level stay local because the field keeps returning the distinctions a reader dropped, as far as the moves can show them.

9. Gravity and entropy: proposals and their exact shadows

Dean proposes two physical consequences. Neither is derived here as physics. Each has an exact mathematical shadow in this paper, and each needs an explicit law before it can be tested against established physics.

9.1 Frozen readers

In the one-state field every reader is frozen: one class, nothing read (Table 7, first row). Working Paper 1 showed that field to be inadmissible, since its only continuation is itself [1]. Any reader that reads anything

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needs at least two states, and the smallest field with two is the swap. A world of readers that never move would be a world of one-state fields.

9.2 A pull into a larger system

Proposal (Dean). Gravity is a pull into a more complex system; without it every reader would be frozen in place.

Shadow (proved here). Three statements of this paper point the same way. A field cannot fall below two states without freezing every reader (§9.1). A move that is one-to-one but not onto stays reversible only on a carrier that grows, $V \subsetneq sVs^{-1} \subsetneq \ldots$ (Proposition 7.3): keeping the move a transformation, with an inverse, pulls the field into a larger one. And every field sits inside larger fields, up to one that contains them all (§7.4).

What a law still needs. Variables, an update rule, units, and a measurable consequence that distinguishes it from established gravitation. Established gravitational physics does not by itself say that gravity drives systems toward greater complexity, and nothing here supplies that step.

9.3 Entropy as what a reader leaves out

Proposal (Dean). Entropy is the reader's incompleteness: what the reader cannot distinguish, not disorder in the field.

Shadow (established). Boltzmann's entropy $S = k \ln W$ counts the microstates W compatible with a macrostate [21], and a macrostate is exactly a reader's class. Table 7 gives $\log_2 W$ for eight readers of the witness: 4.585 bits for the reader that gets nothing back, 0 for the field's own reading, and every value between. On this definition entropy belongs to a reader by construction; the field, read faithfully, has $W = 1$.

What a law still needs. Which readers physical processes implement, and how their classes evolve under the dynamics. Identifying thermodynamic entropy with reader incompleteness in general — beyond Boltzmann's coarse-graining, which is already a reader — is the open step.

9.4 Loose for the reader, exact in the field

Dean's "at human level it's loose, in that we can make mistakes as a reader, but the field never does" is now three exact statements. A reader can merge states that the next move separates (Table 7). A reading can disagree with the field, and its error is the stored lines read (Proposition 5.1). And a reader can fail to decide what it can always read (§5.5). In each case the field's own reading is faithful and its identities hold exactly; every looseness is located in a reader.

10. Corrections and discrepancies

Each correction below closed a loop in a reader (§8.2). Corrections from independent checks are attributed to them; the rest are the writers' own.

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was	is	found by	where
the third stored line of the witness is the octagon $(u \cdot CuC^{-1})^2$	the octagon closes the field, but the shortest third line is the hexagon $(CT^2)^2$; the hole fixes its symmetry	exhaustive search to length 8, guided by the hole's channels	§4.4; ledger §93
BBP: the numerator cancels six of the eight poles	four cancel; $\pm\sqrt{2}$ and $1 \pm i$ survive	sympy factorization	§6.4; [3]
four lines hold the unbounded tape	four lines hold a larger field that contains it; the carrier grows so that s has an inverse	independent check	§7.2; ledger §93
the closed field's extra dimensions are lines stored twice	they are dependencies among boundaries, which can involve several cells	independent check	§4.1
no path returns	no nonempty reduced path returns	independent check	§2.1
the field is all there at radius 4	radius 4 is the diameter for this generating set; motion does not stop	independent check	§7.5
the hole's filter: stabilizer acts trivially in 2^- and fixes a vector in 3a	in general the stabilizer may reverse a face's orientation, and the multiplicity is $\langle \mathrm{Res}\, V, \sigma \rangle$; the hexagon's stabilizer preserves orientation, so the filter applies to it	independent check	§4.4
each witness line is necessary because maps onto $\mathbb{Z}$ kill the others	true for the octagon form; for the hexagon form the certificates are finite-index subgroups with a copy of $\mathbb{Z}$	own, in writing §3	Prop. 3.3
SHA: 2^{256} states per block, as if blocks were a bijection to digests	2^{256} working states; rounds are bijections with their word; absorption and feed-forward merge	independent check	§6.3
$t^8 = 16$ is the BBP construction	$t^8 = 16$ is its carrier line; the π extraction needs the numerator as well	independent check	§6.4
(ledger §90) two thirds of readers have no slide address	true of the symmetric glue only; the alternating glue matches every slide to one reader	own	[2, §90]
(ledger §84) 23 distinct multiplicities at $n = 24$, $k = 6$	13; double precision faked the count; replaced by an integer certificate	own	[2, §86]

Table 9. Corrections, labelled.

11. Open problems

- **The hole names the line, at scale.** Read the minimal missing lines of the family arenas from the channels of their open parts, as §4.4 did for the witness, and check whether the shortest closing shapes are forced for every n.
- **Face quads on the arena.** Table 6 closes the graph-level flow with one prism per marked face. The arena-level version — squares, rotation triangles and the face quads' lifts — is not yet run.
- **Integral and group closure for the family.** The exact line is rational. The $\mathbb{Z}/2$ of the exchange shows that integral information sits beside it; a group-level statement (when the stored shapes present the arena's fundamental groupoid) is open.

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- **SHA's two closures.** A round is a word in two closures of the same 32-bit word, XOR cells and the ADD odometer, and the carry is their disagreement. A reader of that disagreement — which channels the carry feeds, round by round — is the next measurement; no claim about inverting the digest is made.
- **Base change as a change of closure.** Standing item 1 of the programme, the hex transformation space, is untouched here. In the language of this paper a base is an odometer closure of a string of cells; whether base transformations are maps between closures of one set of states is a question, not a result.
- **Laws for the proposals.** Gravity as a pull into a larger system and entropy as what a reader leaves out (§9) each need variables, an update rule, units and a measured consequence; and a proposed physical interaction rule needs progress and preservation (§5.6) proved from the rule itself.
- **The exchange as a field.** §8.2 reads corrections as stored lines closing a reader's loops. Measuring that — the open part of a reader's model before and after a correction — would turn the mirror from an instance into a computation.

12. Methods and reproducibility

All computations are Python 3.11.15 with SymPy 1.14.0 [22] and NumPy 2.4.4; coset enumeration is SymPy's Todd–Coxeter implementation, ranks and Smith forms are exact. Scripts and their outputs are in runs/; each runs as python -I script.py (arguments as listed). Uploaded material from independent checks was copied into separate directories and read only as data. Scripts (first 16 hex digits of SHA-256):

script	SHA-256 prefix	what it computes	sections
presentation.py	21d1afae69e3b4b6	three lines regenerate the witness; readers into $\mathbb{Z}/2$, $\mathbb{Z}/3$, $\mathbb{Z}/6$	3.2, 5.1
witness_shapes.py	8e3d66fd8621edfc	fill table (octagon form); new lines; the count reader	3.6, 4.2, 5.2
witness_shapes_hex.py	6b7d37fc32874b63	fill table, hexagon form	4.2
family_stored.py	67120bbe48d3508b	five lines for Γ_n; the witness at $n = 2$	3.4
entire_field.py	e2858143c52a2905	tape; k-cell tapes; the s induction; paths against states; the third-line search	3.1, 4.4, 7
hex_infinite.py, hex_drop1.py	1a8636149d9eb843, 2b223bf82a52afba	each line needed: certificates and bounds	3.2
mirror.py (argument: the two-cell CSV)	981170f35c0a4925	carry wire; hole from the kernel; cyclic tape; readers; Hom symmetry; total reader; refinement	4.3, 5, 6, 7.3, 8
allchannels.py, allchannels2.py (arguments: fast38.txt fast930.txt fast24.txt)	b1f54201f6830982, a43d62676c8d16dc	every channel of the family's modules	4.5, App. B
face_quads.py	b879db61213e9c3d	face quads	4.5
bbp_poles.py	69156e4f8afb40d8	BBP poles	6.4
sha_round_inverse.py	f680402421bc2deb	SHA-256 rounds inverted	6.3
two_closures.py	6b20a20c4a9aab9e	carry identity; rotation against $\oplus$ and +	6.3

Table 10. Scripts. allchannels.py, allchannels2.py and face_quads.py load the family code of the ledger (gn_fast.py, n4.py, scale.py) [2, §§82–83].

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Appendix A. Proofs

A.1 The witness from three lines (Proposition 3.2)

Let $u = C^{-1}T$ and $u_i = C^i u C^{-i}$, so $T = Cu$, $u_3 = u_0$ by $C^3 = 1$, and each $u_i^2 = 1$. Then $CT^2 = C{\cdot}Cu{\cdot}Cu = (C^2uC^{-2}){\cdot}C^3{\cdot}u = u_2u_0$, and $(CT^2)^2 = 1$ gives $u_2u_0 = (u_2u_0)^{-1} = u_0u_2$. Conjugating by C gives $u_0u_1 = u_1u_0$ and $u_1u_2 = u_2u_1$. So $N = \langle u_0, u_1, u_2\rangle$ is elementary abelian of order at most 8 and normalized by C (which sends u_i to u_{i+1}), and the presented group $N\langle C\rangle$ has at most 24 elements, each of the form $u_0^a u_1^b u_2^c C^r$. The register machine satisfies the three lines (each

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acts as the identity on all 24 states, computed), so by von Dyck the presented group maps onto the register group, which acts simply transitively on 24 states. Both have 24 elements, so the map is an isomorphism. ■

A.2 Five lines for every n (Proposition 3.4)

Conjugating $x^n = 1$ by r gives $(xy)^n = 1$; since x and y commute, $y^n = 1$. The relations $rxr^{-1} = xy$ and $ryr^{-1} = x^{-1}$ make $\langle x, y\rangle$ normal, so every element is $x^a y^b r^j$ with $0 \le a, b < n$, $0 \le j < 6$. On exponent columns, conjugation by r acts by the matrix $M = [[1, -1], [1, 0]]$, whose characteristic polynomial is $\lambda^2 - \lambda + 1$, so $M^6 = I$; the semidirect product $(\mathbb{Z}/n)^2 \rtimes \langle M\rangle$ satisfies the five lines and has $6n^2$ elements. Hence the presented group has exactly $6n^2$ elements and is that product, which is $(\mathbb{Z}[\omega]/n) \rtimes \mu_6 = G_n$ [2, §83]. ■

A.3 The exact line (§4.1)

For a finite 2-complex with a cellular G-action, let $C_2 \to C_1 \to C_0$ be its rational chain complex. Put $A = \ker \partial_1$, $H_2 = \ker \partial_2$ and $D = A / \operatorname{im} \partial_2$. The sequence $0 \to H_2 \to C_2 \to A \to D \to 0$ is exact by these definitions, and every map is G-equivariant, so by Maschke's theorem the characters satisfy $[A] = [C_2] - [H_2] + [D]$ channel by channel. For a connected graph $\operatorname{im} \partial_1$ is the augmentation kernel, so $[A] = [C_1] - [C_0] + [1]$. Signed permutation characters give each term from fixed cells, which is how §4.5's traces were computed [2, §§81, 91]. ■

A.4 The hole from the kernel (Proposition 4.2)

$P = \langle C, u \mid C^3, u^2\rangle = \mathbb{Z}/3 * \mathbb{Z}/2$ acts on its Bass–Serre tree with vertex stabilizers the conjugates of $\langle C\rangle$ and $\langle u\rangle$ and trivial edge stabilizers [7]. The images of C and u in the witness have orders exactly 3 and 2, so $K = \ker(P \to G)$ meets every conjugate of either factor trivially and acts freely; K is the fundamental group of the quotient graph, which has the 24 elements of G as edges and the 8 + 12 cosets of $\langle C\rangle$ and $\langle u\rangle$ as vertices, so K is free of rank 5. The same freeness holds on the Cayley complex of P (vertices and edges are elements of P, and a deduplicated face is fixed only by an element of a conjugate of $\langle C\rangle$ or $\langle u\rangle$); that complex is simply connected, and its quotient by K is the witness field with the triangles and squares filled. So that complex has fundamental group K, and $D = H_1(\cdot; \mathbb{Q}) = K^{ab} \otimes \mathbb{Q}$. The G-equivariant chain complex of the quotient graph, $0 \to D \to \mathbb{Q}G \to \mathbb{Q}G/\langle C\rangle \oplus \mathbb{Q}G/\langle u\rangle \to \mathbb{Q} \to 0$, gives the character identity, which evaluates to $2^- \oplus 3a$. ■

A.5 The error of a reading (Proposition 5.1)

Let $\phi\colon F \to B$ be a homomorphism to an abelian group and $N = \langle\langle R\rangle\rangle$. Two words w, w' reach the same state exactly when $ww'^{-1} \in N$, and then $\phi(w) - \phi(w') = \phi(ww'^{-1}) \in \phi(N)$. Since B is abelian, conjugates have the same image, so $\phi(N)$ is generated by $\phi(R)$. For the count reader the images are (3, 0), (−2, 2) and (−4, 4) — or (2, 4) for the hexagon — and (3, 0) and (−2, 2) already span $\{(a, b) : b \text{ even}, 3 \mid a + b\}$, with determinant 6; the other two vectors lie in it. ■

A.6 Reading from both sides (§5.4)

For finite-dimensional rational or complex G-modules, $\dim \mathrm{Hom}_G(V, W) = \langle \chi_V, \chi_W\rangle = |G|^{-1} \Sigma_g \chi_V(g)\, \chi_W(g)^*$. That number is a nonnegative integer, hence equal to its complex conjugate $\langle \chi_W, \chi_V\rangle = \dim \mathrm{Hom}_G(W, V)$. ■

A.7 One commutation becomes all of them (Proposition 7.2)

From $a^s = aa^t$ and $[s, t] = 1$, conjugating by t^i gives $b_i^s = b_i b_{i+1}$. Suppose all lamps at separation at most k commute (true for $k = 1$ from $[a, a^t]$ and translation by t). Conjugating $[b_0, b_k] = 1$ by s gives $xyzw = zwxy$ with $x = b_0$, $y = b_1$, $z = b_k$, $w = b_{k+1}$. The pairs (x, z), (y, z) and (y, w) are at separation at most k, so $xyzw = xzyw = zxyw = zxwy$; hence $zxwy = zwxy$ and $xw = wx$, that is $[b_0, b_{k+1}] = 1$. Translating by t gives every pair at separation $k + 1$, and

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induction gives all. That the lamps are moreover independent — the subgroup is the lamplighter group and not a quotient of it — is the embedding theorem of [16]. ■

A.8 The carrier has to grow (Proposition 7.3)

Identify finite-support lamp configurations with $V = \mathbb{F}_2[t, t^{-1}]$ and the move with multiplication by $S = 1 + t$. If $Sx = 0$ then $x_i = x_{i-1}$ for every i, and finite support forces $x = 0$: S is one-to-one. The coordinate sum of Sx is twice that of x, so every image has even parity; conversely an even-parity y of finite support is S of its partial sums $x_i = \Sigma_{j \le i}\, y_j$, which also have finite support. So the image is the even part of V. With $b^s = s^{-1}bs$, this says $s^{-1}Vs$ is the even part, a proper subgroup of V; conjugating by s, $V \subsetneq sVs^{-1}$, and repeating gives the strictly increasing chain. On its union $\mathbb{F}_2[t, t^{-1}, (1 + t)^{-1}]$ the move is invertible. ■

A.9 The ring splits (Proposition 7.4)

With $d = 2^\nu$, in characteristic 2 $t^k - 1 = (t^q - 1)^d$, and $t^q - 1 = (t + 1)h_q$ with $h_q(1) = q \equiv 1$, so $(t + 1)$ and h_q are coprime and the Chinese remainder theorem splits $\mathbb{F}_2[t]/(t^k - 1)$ into $\mathbb{F}_2[\varepsilon]/(\varepsilon^d)$ and $\mathbb{F}_2[t]/(h_q{}^d)$, $\varepsilon = 1 + t$. On the first summand, of dimension d, multiplication by ε^m has rank $d - \min(m, d)$; on the second, of dimension $k - d$, $1 + t$ is a unit. Adding, rank $S^m = k - \min(m, d)$. When $k = d$ the second summand vanishes, the ring is $\mathbb{F}_2[\varepsilon]/(\varepsilon^k)$, and an element is a unit exactly when it is not divisible by ε, that is when its value at $t = 1$ is 1 — a sum of distinct rotations with an odd number of terms. ■

Appendix B. Every channel of the family's modules

For the arena of two tokens on Γ_n with commuting squares as the stored shapes, $n \ge 3$, multiplicity per complex irreducible of G_n [2, §§83, 91, 92]. Braces give {n odd, n even}; $\delta = (-1)^{n/2}$; [d] = 1 on a double channel (n even, address in $2\mathbb{Z}[\omega]$); 3a and 3b exist for even n, 2c and 2r for $3 \mid n$. The square module is $C_2 = H_2 + A - D$ in every channel.

channel	open D	bubbles H_2	loops A
1^+	3	$11n^2 - 21n/2 + \{1/2, -1\}$	$29n^2/2 + 29n/2 - 1$
2^+	2	$11n^2 - 21n/2 + \{1/2, -1\}$	$29n^2/2 + 29n/2 - 2$
$1^-, 2^-$	1, 1	$13n^2 - 14n + \{3, 2\}$	$31n^2/2 + 31n/2 - \{10, 14\}$
3a	5	$36n^2 - 155n/4 + 23/4 - 11\delta/4$	$45n^2 + 45n - (49 + 7\delta)/2$
3b	6	$36n^2 - 143n/4 + 11/4 - 11\delta/4$	$45n^2 + 45n - (25 + 7\delta)/2$
2c, 2r	4	$24n^2 - 25n + 4 - 3[n \text{ even}]$	$30n^2 + 30n - 11 - 3[n \text{ even}]$
6g (free, generic)	12	$72n^2 - 75n + 12 + [n \text{ even}](3 - 12[\mathrm{d}])$	$\{90n^2 + 90n - 33, 90n^2 + 90n - 30 - 12[\mathrm{d}]\}$
6L (free, on a line)	11	$6g + (n - 1 + \varepsilon)/2$, $\varepsilon = \{0, -1 + 2[\mathrm{d}]\}$	$6g - \{1, 2[\mathrm{d}]\}$

Table 11. Closed forms. All agree with the 182 computed channel rows, n = 3..22 and 24, in every module (allchannels.py, allchannels2.py). The 3a and 3b bubble forms are §91's sum $72n^2 - 149n/2 + 17/2 - 11\delta/2$ split by the difference $3n - 3$.

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